Reproducibility in Data Science: A Data Ethics Project

Background and Context

Reproducibility Crisis in Data Science

Scientific reproducibility is fundamental to research integrity and the advancement of knowledge. However, many published results in data science and machine learning prove difficult or impossible to reproduce. This "reproducibility crisis" raises significant ethical concerns about: - Scientific integrity and trust - Research transparency - Equal access to computational resources - Barriers to scientific verification - Impact on real-world applications.

The focus of this project is to have your group reproduce the results of a data science paper. You may or may not succeed in this task, but you will be able to document the steps you took and the barriers you encountered in achieving this goal.

Project Overview

This project examines reproducibility in data science by attempting to replicate results from a published paper. Groups will document challenges, variations in results, and missing information while considering the ethical implications of reproducibility issues.

Project Goals

- Attempt to reproduce key results from the selected paper
- Document all challenges and deviations
- Analyze implications for scientific integrity
- Propose best practices for reproducible research
- Create a fully reproducible version of your work

Paper Selection Criteria

The selected paper must have:

- Publicly available dataset(s)
- Clear methodology description
- Multiple analysis components that can be divided among team members
- Published within the last 5 years
- Significant data science or statistical analysis component

Paper Selection Resources

Where to Find Papers

1. **Research Repositories**

  - arXiv.org (particularly cs.LG and stat.ML sections)
  - Papers With Code (paperswithcode.com)
  - ACM Digital Library
  - IEEE Xplore
  - Google Scholar
- **Recommended Paper Types**
  - Predictive modeling on public datasets
  - Natural language processing analyses
  - Time series forecasting
  - Image classification studies
  - Public health data analysis
  - Social media analytics
  - Climate data analysis
  - Economic data studies
- **Example Papers and Datasets**
  - UCI Machine Learning Repository analyses
  - Kaggle competition winner write-ups
  - MNIST/CIFAR classification studies
  - COVID-19 data analyses
  - Census data studies
  - Twitter sentiment analyses
  - Financial market predictions

Project Components

1. Initial Analysis

- Detailed reading and documentation of the paper
- Identification of key results to reproduce
- Assessment of available data and code
- Planning reproduction strategy
- Division of reproduction tasks

2. Reproduction Attempt

- Data collection and preprocessing
- Implementation of methods
- Recreation of key visualizations
- Documentation of all steps
- Comparison with original results

3. Ethics Analysis

- Impact of reproducibility challenges
- Resource and access considerations
- Scientific communication issues
- Implications for stakeholders
- Recommendations for improvement

Key Checkpoints

- **End of First Weekend**
  - Paper selected and analyzed
  - Project plan created
  - Repository set up
  - Initial documentation complete
- **End of First Week**
  - Data collected and processed
  - Initial reproduction attempts started
  - Clear understanding of challenges
  - Methods documentation begun
- **End of Second Weekend**
  - Major reproduction attempts completed
  - Analysis of differences started
  - Report drafting underway
  - Ethics discussion ongoing
- **Final Submission (Wednesday Midnight)**
  - Complete reproduction documentation
  - Final report
  - All code and data
  - Meeting summaries
  - Individual reflections

Evaluation Notes

Success Criteria for Reproducing Results (Note the notes on grading below)

1. **Complete Success**: Results match within reasonable statistical variance
2. **Partial Success**: General trends reproduced but specific numbers differ
3. **Unable to Reproduce**: Clear documentation of:
   - Missing information
   - Resource constraints
   - Implementation challenges

  - Potential methodological issues
  - Possible explanations for differences

What Will Not Affect Grades

- Inability to exactly match original results
- Different numerical outcomes with documented reasons
- Resource constraints preventing full implementation
- Missing information in original paper
- Software version differences causing variations

What Will Affect Grades

- Quality of documentation
- Thoroughness of attempt
- Analysis of challenges
- Ethical implications discussion
- Team collaboration
- Technical implementation quality
- Clear communication of findings

Additional Considerations

Resource Management

- Start with smaller components
- Test on subset of data first
- Use free compute resources (Google Colab, etc.)
- Document resource limitations

Common Challenges

- Missing hyperparameters
- Undefined data preprocessing steps
- Unstated assumptions
- Version conflicts
- Random seed issues
- Hardware dependencies
- Data availability changes

Project Steps

Step 1: Set Up Project Infrastructure

- **Create GitHub Repository**
  - Name: `DATA6550-Reproducibility`
  - Required directory structure:

```
final_report.pdf
readme.md
Code/
  Lastname1/
  Lastname2/
  Lastname3/
  Lastname4/
Data/
  original/
  processed/
Collaboration/
  WeekA.pdf
  WeekB.pdf
Analysis/
  figures/
  results/
Environment/
  requirements.txt
```

- **Set Up Shared Document**
  - Create a shared Google document for the final report of this project. Google Docs will automatically record all document updates and it is one of the ways the instructor will track your contributions. Share the document with the team, the instructor, and add a link to it in the Dropbox for the project in D2L along with uploading a PDF of the document. My Gmail account is jfwallin@gmail.com when you add me to the document (with editing rights, please). Please submit only one PDF and link per group
- Choose a communication platform (D2L forums, Teams, Zoom, etc.)
  - Invite all group members and instructor to the platform
  - Establish regular check-in schedule
  - Create AI summaries of the meetings
  - Document communication protocols in repository README

Step 2: Paper Analysis

- Select paper meeting criteria
- Document key findings to reproduce

- Assess data and code availability
- Create reproduction plan
- Divide responsibilities

Step 3: Implementation

- Create reproducible environment
- Collect and preprocess data
- Implement analysis methods
- Create visualizations
- Document all steps

Step 4: Analysis and Documentation

- Compare results with original
- Document discrepancies
- Identify missing information
- Analyze sources of variation
- Create comprehensive documentation

Deliverables

GitHub Repository

- Complete reproduction code
- Environment configuration files
- Documentation of all steps
- Comparison with original results
- Collaboration summaries (**WeekA.pdf** and **WeekB.pdf**) and supplemental document if required

Final Report (5 pages) – final_report.pdf and Google Doc Link

- Introduction
  - Paper summary
  - Reproduction goals
- Methodology
  - Reproduction approach
  - Environmental setup
  - Data processing steps
- Results
  - Comparison with original
  - Discrepancy analysis
- Ethics Discussion
  - Reproducibility challenges

  - Resource considerations
  - Scientific integrity implications
- Recommendations
  - Best practices
  - Process improvements

Collaboration summaries

- (**WeekA.pdf** and **WeekB.pdf**) to GitHub and the Dropbox.

Personal Reflections

- Upload the personal reflection document of YOUR contributions to the individual assignment Dropbox. This need only be a paragraph or two of your contributions and the lessons you learned in the project. Your individual contributions will be evaluated based on this document, your commits to GitHub, the summaries of the discussions, and your work on the Google document as found through track changes.

Presentation (15 minutes)

1. Paper overview
2. Reproduction methodology
3. Results comparison
4. Challenges encountered
5. Ethical implications
6. Recommendations

Technical Requirements

1. Use version control for all code
2. Document computational environment
3. Create reproducible workflows
4. Include all data processing steps
5. Provide clear installation instructions

Evaluation Criteria

Individual (40%)

- Code quality and documentation
- GitHub contributions
- Participation in discussions
- Understanding of reproducibility challenges

Group (60%)

- Success in reproduction attempts
- Quality of documentation
- Analysis of ethical implications
- Presentation effectiveness
- Repository organization

Tips for Success

- Start with simple reproductions first
- Document everything, including “failed” attempts
- Consider computational resource requirements
- Maintain clear communication about challenges
- Focus on both technical and ethical aspects